

Supplementary Material for: Adaptive Neuromorphic Control for Rhythmic Pendulum Oscillation

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1. OVERVIEW

This document contains supplementary figures, extended mathematical derivations, and additional simulation plots that accompany the main manuscript.

2. PROOF OF THEOREM 1

Proof. To demonstrate the temporal scaling property, we introduce the non-dimensional time variable $\tilde{t} = t/\tau_f$. By the chain rule, the time derivative transforms as $\frac{d}{dt} = \frac{d\tilde{t}}{dt} \frac{d}{d\tilde{t}} = \frac{1}{\tau_f} \frac{d}{d\tilde{t}}$. Substituting this transformation into (S9) yields the rescaled dynamics, given by

$$\begin{aligned} \frac{dv_i}{d\tilde{t}} &= -v_i + g_f^- \tanh(v_i) - g_s^+ \tanh(v_{s,i}) + \\ &\quad g_s^- \tanh(v_{s,i} + 0.9) - g_{us}^+ \tanh(v_{us,i} + 0.9) + \\ &\quad I_{\text{syn},i}(\tilde{t}) + I_i(\tilde{t}), \\ \frac{dv_{s,i}}{d\tilde{t}} &= k_s(v_i - v_{s,i}), \\ \frac{dv_{us,i}}{d\tilde{t}} &= k_{us}(v_i - v_{us,i}), \end{aligned}$$

where $k_s = \tau_f/\tau_s$ and $k_{us} = \tau_f/\tau_{us}$.

Defining the state vector $\mathbf{v} = [v_i, v_{s,i}, v_{us,i}]^T$, the non-dimensional system is:

$$\frac{d\mathbf{v}}{d\tilde{t}} = \mathbf{G}(\mathbf{v}, k_s, k_{us}, \text{gains}, I_i(\tilde{t}), I_{\text{syn},i}(\tilde{t})). \quad (\text{S1})$$

Since the time constant ratios k_s and k_{us} are held constant, the vector field $\mathbf{G}$ is entirely independent of the absolute value of τ_f . Consequently, the geometric structure of all trajectories in the phase space defined by $\frac{d\mathbf{v}}{d\tilde{t}}$ is invariant under changes to τ_f .

The physical time t is recovered via $t = \tau_f \tilde{t}$. If τ_f is scaled by a factor α , such that $\tau_f' = \alpha \tau_f$, the time required to traverse the invariant non-dimensional trajectory $\tilde{T}$ is $T' = \tau_f' \tilde{T} = \alpha \tau_f \tilde{T} = \alpha T$. Thus, the period T scales linearly with τ_f . This concludes the proof.

3. PROOF OF PROPOSITION 1

Proof. The proof proceeds via a fast-slow decomposition and fixed-point analysis of the one-dimensional fast subsystem through 3 steps.

Step 1. Fixed Point and Nullcline Analysis

By treating the slow state variables and external inputs as a constant parameter $C \in \mathbb{R}$, the fixed points, v_i^* , are determined by setting $\dot{v}_i = 0$, yielding the algebraic equation:

$$v_i^* = g_f^- \tanh(v_i^*) + C. \quad (\text{S2})$$

The number of fixed points corresponds to the number of intersections between the identity line $y_1 = v_i$ and the fast nullcline $y_2 = g_f^- \tanh(v_i) + C$. C acts as a vertical translation of y_2 , while the maximum slope of y_2 is governed solely by g_f^- , specifically $\max\left(\frac{\partial y_2}{\partial v_i}\right) = g_f^-$.

Step 2. Stability and Bifurcation Condition

The local stability of a fixed point v_i^* is determined by the sign of the Jacobian, $J = \partial \dot{v}_i / \partial v_i$, evaluated at v_i^* :

$$J = \left(-1 + g_f^- \text{sech}^2(v_i^*)\right) / \tau_f. \quad (\text{S3})$$

A fixed point is stable if $J < 0$ and unstable if $J > 0$. A saddle-node bifurcation occurs when the fixed point becomes marginally stable ($J = 0$), which requires the critical potential v_i^{crit} to satisfy the condition:

$$g_f^- \text{sech}^2(v_i^{\text{crit}}) = 1. \quad (\text{S4})$$

Step 3. Bifurcation Analysis by Case

The potential for multi-stability hinges on whether the maximum slope of the fast nullcline exceeds the slope of the identity line, i.e., whether $g_f^- > 1$. We have two cases:

(1) Case $g_f^- \leq 1$ (Monostable Regime): Since $g_f^- \leq 1$ and $\text{sech}^2(v_i^*) \leq 1$, it follows that $g_f^- \text{sech}^2(v_i^*) < 1$ for all $v_i^* \in \mathbb{R}$, which implies $J < 0$. Therefore, any fixed point must be stable. Geometrically, the maximum slope of y_2 is less than or equal to 1, guaranteeing a single intersection with y_1 for all $C \in \mathbb{R}$. The system is globally monostable.

(2) Case $g_f^- > 1$ (Bistable Regime): The maximum slope of y_2 is $g_f^- > 1$. This allows for three intersections

with the identity line y_1 when the vertical shift C is sufficiently small. These three fixed points consist of two stable nodes ($J < 0$) separated by one central unstable saddle ($J > 0$). The boundaries of this bistable region are defined by the saddle-node bifurcation, where two fixed points merge and vanish. This occurs when the fixed point condition (S2) and the marginal stability condition (S4) are simultaneously satisfied for the critical values $(v_i^{\text{crit}}, C_{\text{crit}})$. From (S4), the critical potential is:

$$v_i^{\text{crit}} = \text{arsech}\left(1/\sqrt{g_f^-}\right). \quad (\text{S5})$$

Substituting (S5) into (S2), we have

$$C_{\text{crit}}(g_f^-) = \left|v_i^{\text{crit}} - g_f^- \tanh(v_i^{\text{crit}})\right|. \quad (\text{S6})$$

Bistability is possible if and only if $g_f^- > 1$ and $|C| < C_{\text{crit}}(g_f^-)$. If $|C| \geq C_{\text{crit}}(g_f^-)$, the nullcline is shifted such that only one stable fixed point remains. The critical gain $g_f^- = 1$ marks the pitchfork bifurcation point for the emergence of the bistable regime. This concludes the proof.

4. PROOF OF PROPOSITION 2

Proof. From (S9), The v_i -nullcline defined by the condition $\dot{v}_i = 0$ is given by

$$v_i = g_f^- \tanh(v_i) + \Phi + I_{\text{ext}} = \mathcal{F}(v_i, v_{s,i}, v_{us,i}; \cdot),$$

where $I_{\text{ext}} = I_{\text{syn},i} + I_i$, and Φ encapsulates all slow-variable contributions defined as

$$\Phi = -g_s^+ \tanh(v_{s,i}) + g_s^- \tanh(v_{s,i} + 0.9) - g_{us}^+ \tanh(v_{us,i} + 0.9). \quad (\text{S7})$$

Because the term $g_f^- \tanh(v_i)$ operates independently of the synaptic gains, the translation of the nullcline $\mathcal{N}$ along the $\mathcal{F}$ -axis is governed entirely by the partial derivatives of the modulating term Φ . The specific effects of varying each respective gain are as follows:

(1) Slow Inhibitory Gain (g_s^+) The derivative is: $\frac{\partial \Phi}{\partial g_s^+} = -\tanh(v_{s,i})$. For active states where $v_{s,i} \geq 0$, we have $\tanh(v_{s,i}) \geq 0$, which ensures $\frac{\partial \Phi}{\partial g_s^+} \leq 0$. Thus, an increase in g_s^+ induces a negative (downward) translation of $\mathcal{N}$, promoting inhibition.

(2) Slow Excitatory Gain (g_s^-) The derivative is: $\frac{\partial \Phi}{\partial g_s^-} = \tanh(v_{s,i} + 0.9)$. The positive offset 0.9 ensures that $\tanh(v_{s,i} + 0.9) > 0$ for all physiologically relevant $v_{s,i}$. Hence, $\frac{\partial \Phi}{\partial g_s^-} > 0$. An increase in g_s^- produces a positive (upward) translation, promoting excitation.

(3) Ultra-Slow Adaptive Gain (g_{us}^+) The derivative is: $\frac{\partial \Phi}{\partial g_{us}^+} = -\tanh(v_{us,i} + 0.9)$. Since $\tanh(v_{us,i} + 0.9) > 0$, we have $\frac{\partial \Phi}{\partial g_{us}^+} < 0$. Increases in g_{us}^+ induce a negative (downward) nullcline shift.

5. PROOF OF PROPOSITION 3

Proof. The v_i -nullcline is determined by the condition $\dot{v}_i = 0$, which yields the explicit relationship:

$$v_i = g_f^- \tanh(v_i) + \Phi(\mathbf{v}_{\text{slow}}) + I_i(t), \quad (\text{S8})$$

where $\Phi(\mathbf{v}_{\text{slow}})$ represents the constant contribution of the slow variables (as defined in Theorem 2, and $\mathbf{v}_{\text{slow}} =$

$(v_{s,i}, v_{us,i})$). The fixed points occur at the intersections of the nullcline (S8) and the identity line, $y = v_i$.

Since the external input $I_i(t)$ is an additive term independent of the state variables $\mathbf{v}$, any change in $I_i(t)$ results in a pure vertical translation of the entire nullcline within the phase portrait. This translation directly modulates the overall constant bias $C = \Phi(\mathbf{v}_{\text{slow}}) + I_i(t)$.

As $I_i(t)$ increases, the resultant upward translation of the nullcline forces the low-voltage stable and central unstable fixed points to merge. Once $I_i(t)$ exceeds a critical value, these two fixed points annihilate via a saddle-node bifurcation on a fold. This event removes the stable attractor associated with the silent regime. Consequently, the system is left with only the single remaining stable fixed point located on the upper branch of the nullcline, forcing the trajectory to transition abruptly and irreversibly to the high-voltage active state.

6. CIRCUIT REPRESENTATION OF THE HCO MODEL

The neuromorphic control framework is governed by the state equations detailed in (S9). This system of nonlinear ordinary differential equations models the fast membrane potential $v_i(t)$, alongside the temporal evolution of the slow, $v_{s,i}(t)$, and ultra-slow, $v_{us,i}(t)$, state variables.

$$\begin{cases} \dot{v}_i(t) = \frac{1}{\tau_f} \left(-v_i(t) + g_f^- \tanh(v_i(t)) - g_s^+ \tanh(v_{s,i}(t)) \right. \\ \quad \left. + g_s^- \tanh(v_{s,i}(t) + 0.9) - g_{us}^+ \tanh(v_{us,i}(t) + 0.9) \right. \\ \quad \left. + I_{\text{syn},i}(t) + I_i(t) \right), \\ \dot{v}_{s,i}(t) = \frac{1}{\tau_s} (v_i(t) - v_{s,i}(t)), \\ \dot{v}_{us,i}(t) = \frac{1}{\tau_{us}} (v_i(t) - v_{us,i}(t)), \end{cases} \quad (\text{S9})$$

Figure S1 visualizes the corresponding circuit architecture for this model. The schematic maps the mathematical parameters to physical components, illustrating how the time constants ($\tau_f, \tau_s, \tau_{us}$) are realized via resistor-capacitor networks and how the synaptic gains ($g_f^-, g_s^+, g_s^-, g_{us}^+$) translate into amplifier configurations. This diagram provides a structural mapping of the internal feedback loops, self-excitation pathways, and reciprocal synaptic coupling, $I_{\text{syn},i}(t)$, defined in the analytical formulation.

7. VISUALIZATION OF PROPOSITION 1: NULLCLINE ANALYSIS OF SELF EXCITATION GAIN

To visualize Proposition 1 from the main manuscript, the influence of slow dynamics and external inputs is set to zero ($C = 0$). A non-zero C vertically translates the fast nullcline, shifting the fixed point coordinates while preserving the qualitative stability regimes. Figure S2 illustrates the saddle-node bifurcation structure delineated in Theorem 3. Fixed points correspond to the intersections of the identity line, $y = v_i$, and the fast nullcline, $y = g_f^- \tanh(v_i)$. Three distinct dynamic regimes emerge depending on the self-excitation gain g_f^- :

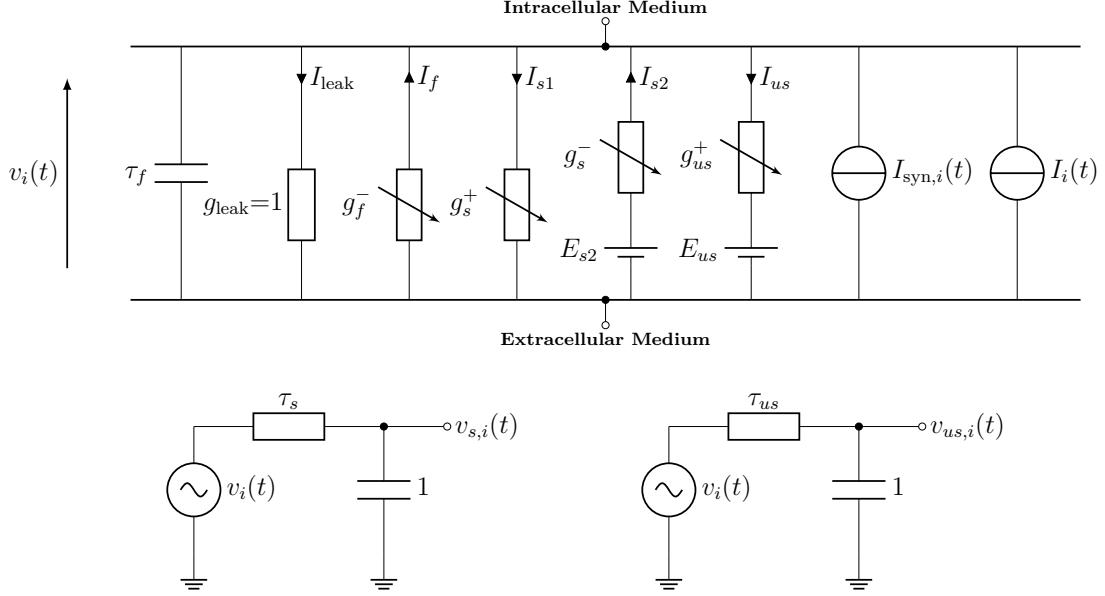


Fig. S1. Equivalent electrical circuit representing the full Half-Center Oscillator model defined in Equation (1) in the main manuscript. The upper circuit models the membrane potential $v_i(t)$, where the membrane capacitance is defined directly by the time constant τ_f . The current arrows explicitly denote the physical direction of charge flow (inward or outward) to perfectly match the positive and negative signs in the mathematical formulation ($\tau_f \dot{v}_i = \sum I_{in} - \sum I_{out}$). Specifically, the outward (downward) currents are the passive decay $I_{leak} = v_i(t)$, the first slow current $I_{s1} = g_s^+ \tanh(v_{s,i}(t))$, and the ultra-slow current $I_{us} = g_{us}^+ \tanh(v_{us,i}(t) + 0.9)$. The inward (upward) currents are the fast current $I_f = g_f^- \tanh(v_i(t))$, the second slow current $I_{s2} = g_s^- \tanh(v_{s,i}(t) + 0.9)$, and the external inputs $I_{syn,i}(t)$ and $I_i(t)$. The voltage offsets inside the non-linear $\tanh$ functions act as reversal potentials, indicated by the batteries $E_{s2} = -0.9$ and $E_{us} = -0.9$. The lower sub-circuits illustrate the first-order RC low-pass filters that generate the kinetic state variables $v_{s,i}(t)$ and $v_{us,i}(t)$, normalized with a unit capacitance.

- (1) **Monostable Regime** ($g_f^- < 1$, Blue): The maximum nullcline slope is less than 1, yielding a single intersection and a unique, globally stable fixed point.
- (2) **Bifurcation Boundary** ($g_f^- = 1$, Magenta): The nullcline is tangent to the identity line at the origin, denoting the transition point for a degenerate saddle-node bifurcation.
- (3) **Bistable Regime** ($g_f^- > 1$, Red): The maximum slope exceeds 1, producing three intersections. The central intersection represents an unstable fixed point (saddle) that separates two stable outer fixed points, allowing the neuron to converge to one of two non-zero activity states.

8. VISUALIZATION OF PROPOSITION 2: SYNAPTIC GAIN MODULATION

To visualize Proposition 2 from the main manuscript, this section demonstrates how induced nullcline translations govern the geometry and stability of the fast subsystem's fixed points. Figure S3 provides a geometric interpretation of Proposition 2, illustrating how synaptic gains modulate the v_i -nullcline structure. These translations enable the network to cycle between the active and quiescent states required for limit cycle oscillations. Three distinct modulatory mechanisms are observed:

- (1) **Inhibitory Modulation** (g_s^+): The transition from the blue ($g_s^+ = 0.5$) to the red ($g_s^+ = 1.5$) curve exhibits a downward vertical shift of the nullcline. This translation lowers the intersection points with

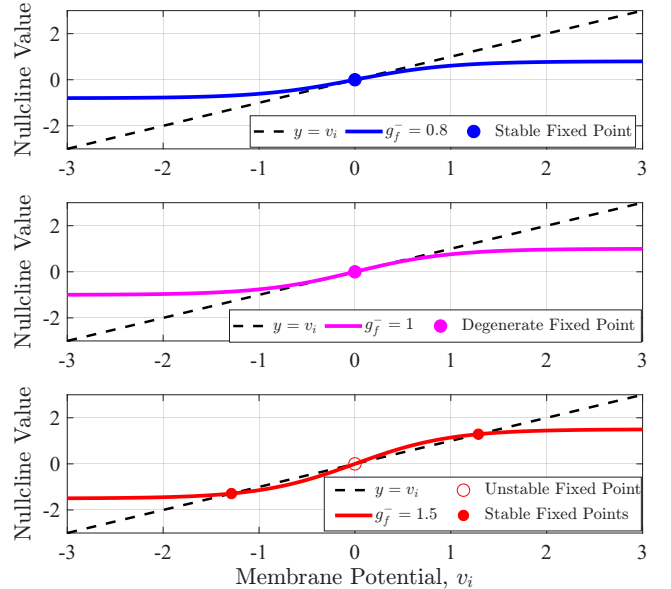


Fig. S2. The v_i -nullcline, $y = g_f^- \tanh(v_i)$, plotted for three self-excitation gain regimes: $g_f^- = 0.8 < 1$, $g_f^- = 1$, and $g_f^- = 1.5 > 1$. Fixed points correspond to the intersections of the nullcline with the dashed identity line, $y = v_i$.

the identity line, $y = v_i$, facilitating transitions to lower-voltage fixed points and stabilizing the quiescent regime.

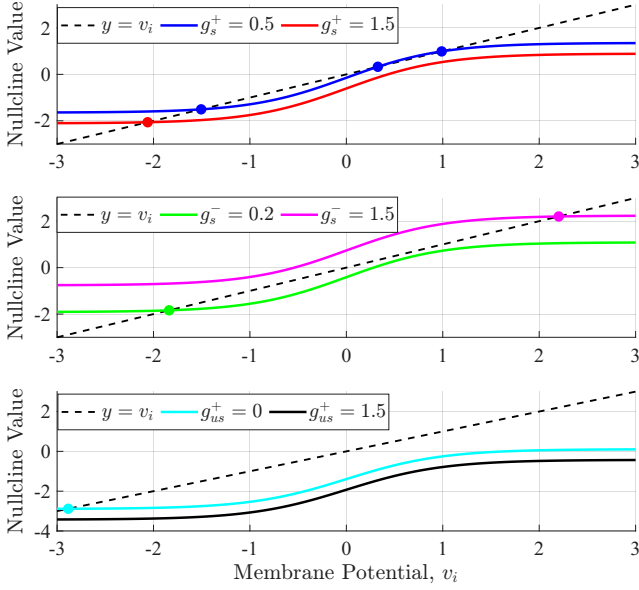


Fig. S3. Geometric influence of synaptic gains on the v_i -nullcline. The intersections between each nullcline and the identity line $y = v_i$ (black dashed line) define the system's fixed points, indicated by solid circular markers. Three modulation mechanisms are illustrated: (Top) variation of the inhibitory gain $g_s^+ \in \{0.5, 1.5\}$ (blue and red curves); (Middle) variation of the excitatory gain $g_s^- \in \{0.2, 1.5\}$ (green and magenta curves); and (Bottom) evolution of the ultra-slow variable $g_{us}^+ \in \{0, 1.5\}$ (cyan and black curves), highlighting adaptive, time-dependent modulation of the nullcline geometry. The self-excitation parameter is fixed at $g_f^- = 1.5$, while all remaining gains are held constant within each respective scenario.

- (2) **Excitatory Modulation** (g_s^-): The progression from the green dashed ($g_s^- = 0.2$) to the magenta dashed ($g_s^- = 1.5$) curve produces an upward vertical shift. This establishes fixed points in the high-voltage region, sustaining active bursting behavior.
- (3) **Adaptive Modulation** (g_{us}^+): The transition from the cyan ($g_{us}^+ = 0$) to the black ($g_{us}^+ = 1.5$) curve demonstrates time-dependent adaptation. As the ultra-slow variable $v_{us,i}(t)$ accumulates during activity, the nullcline drifts downward. This progressive shift destabilizes the high-voltage fixed points, providing a self-terminating mechanism for rhythmic oscillations. The magnitude of g_{us}^+ regulates the adaptation rate, thereby defining burst duration and frequency.

These systematic nullcline transformations collectively regulate the existence and accessibility of fixed points between quiescent and active states.

9. VISUALIZATION OF PROPOSITION 3: BIFURCATION CONTROL VIA EXTERNAL INPUT

Supplementary Fig. S4 illustrates the mechanism of bifurcation control governed by the external input current, $I_i(t)$. Fixed points correspond to the intersections of the v_i -nullcline and the identity line ($y = v_i$). In the unforced state ($I_i = 0$, blue curve), the system exhibits bistability,

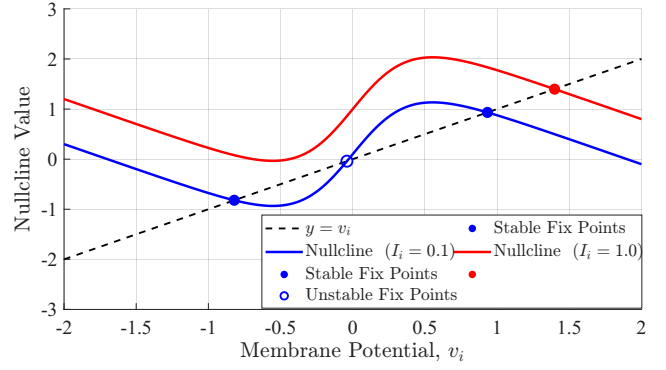


Fig. S4. Effect of the external input current I_i on the v_i -nullcline ($\dot{v}_i = 0$). The solid blue line denotes the nullcline at $I_i = 0$. Applying a positive input ($I_i = 1$) produces a vertical translation, represented by the red line. Fixed points correspond to the intersections with the identity line ($y = v_i$) and are indicated by circles.

characterized by three fixed points: two stable nodes on the upper and lower branches separated by an unstable saddle in the middle.

The application of a positive input current ($I_i = 1$, red curve) induces a vertical translation of the nullcline. This translation eliminates the stable fixed point on the lower branch, driving the state trajectory toward the singular remaining stable fixed point on the upper branch. This geometric transformation initiates the transition to the active (spiking/bursting) state, thereby defining the mechanism for state switching.